

Software Engineering Lab 04

Q*bert Repair Lab — Implementation Conversation Record

Repository: ramneek-01/SE_LAB1_PES1UG24AM220 · Folder: Lab_04 · Branch: lab-04

Scope and accuracy note

This document records the implementation decisions and troubleshooting relevant to this lab. It is a prepared implementation record, not a verbatim export of every chat message. The game could not be executed in the available environment, and no gameplay videos were created. Runtime behavior remains unverified.

Task 1 — Fix pyramid projection

Center pyramid rows by fixing the row-offset arithmetic in `cube_center`. The implementation uses true division for the row offset instead of integer division.

```
Before: (col - row // 2) * CUBE_W
After:  (col - row / 2) * CUBE_W
```

Task 2 — Add level colour palettes

Choose a palette according to the current level, cycling through configured palettes.

```
return palettes[(level - 1) % len(palettes)]
```

Task 3 — Add cube completion flash

Provide brief visual feedback when a cube reaches its target colour. The completion handler records a flash expiry timestamp in `COMPLETION_FLASHES`; drawing code uses it to blend the cube toward white.

```
COMPLETION_FLASHES[cell] = pygame.time.get_ticks() + COMPLETION_FLASH_MS
```

Task 4 — Add bonus-life threshold

Set a score threshold for awarding an extra life; the implementation selects 1,000 points.

```
def bonus_life_threshold():
    return 1000
```

Repository workflow

The `Lab_04` folder was added to the user's repository. Separate commits were created for the four task implementations. A pull request was opened and the folder later appeared on main. An internal `SUBMISSION_STATUS.txt` note was removed, and this PDF was added to the folder.

Troubleshooting record

Git for Windows was installed and the user's repository was cloned successfully. Pygame installation failed under Python 3.14 because pip could not find a compatible binary distribution and the source build failed. Pygame could not be installed in the available execution environment because network access was unavailable.

Final verification status

Source changes and repository files were created remotely. The game was not run successfully, so no claim is made that it has been runtime-tested. The required before and after gameplay recordings remain outstanding. Videos from another repository were not copied or represented as the user's own evidence.